Solutions to Deferred/Supplementary Examination

1.(a) False. The harmonic series is divergent and has only positive terms, so any rearrangement of this series will also diverge. 2

(b) False. The set $\mathbb{N}$ has no limit points. 2

(c) True. If a subset S of $\mathbb{R}$ is uncountable, then there must be an $n \in \mathbb{Z}$ such that $S \cap [n, n+1]$ is uncountable. It is possible to choose a sequence $(x_k)_{k=1}^{\infty}$ of distinct points in $S \cap [n, n+1]$. According to the Bolzano-Weierstrass Theorem, this sequence has a convergent subsequence. The limit of this subsequence will be a limit point of S. 4

(d) False. The power series $\displaystyle\sum_{k=1}^{\infty} \frac{x^k}{k}$ converges when $x = -1$, but not when $x = 1$. 2

2.(a) We use induction. First note that $a_1 = 1$ and $a_2 = 3\frac{1}{16}$, so the inequalities hold in the case when $k = 1$. Suppose now that

$$1 \leqslant a_n \leqslant a_{n+1} \leqslant 6$$

for some $n \in \mathbb{N}$. Then

$$3 \leqslant 3 + \left(\frac{a_k}{4}\right)^2 \leqslant 3 + \left(\frac{a_{n+1}}{4}\right)^2 \leqslant \frac{84}{16},$$

so

$$1 \leqslant 3 \leqslant a_{n+1} \leqslant a_{n+2} \leqslant 5\tfrac{1}{4} \leqslant 6.$$

The Principle of Mathematical Induction tells us that the inequalities hold for all $k \in \mathbb{N}$. 5

(b) The *Monotone Convergence Theorem* says that if a sequence is increasing and bounded above, then it converges (to its supremum). Since we have shown $a_k \leqslant a_{k+1}$ for all $k \in \mathbb{N}$, the sequence $(a_k)_{k=1}^{\infty}$ is increasing. We have also proved that $(a_k)_{k=1}^{\infty}$ is bounded above by 6. 4

(c) Put $x = \lim\limits_{k\to\infty} a_k$. Taking limits on both sides of the equation $a_{k+1} = 3 + \frac{1}{16}a_k^2$, we deduce that

$$\begin{aligned} & x = 3 + \tfrac{1}{16}x^2 \\ \iff & (x-4)(x-12) = 0 \\ \iff & x = 4 \text{ or } x = 12. \end{aligned}$$

Since $a_k \leqslant 6$ for all k, we conclude that $\lim\limits_{k\to\infty} a_k = 4$. 4

3.(a) Put

$$L = \lim_{k\to\infty} \frac{4^{k+1}(k+2)}{(k+1)!} \frac{k!}{4^k(k+1)}.$$

Then $L = \lim\limits_{k\to\infty} \dfrac{4(k+2)}{(k+1)^2} = 0 < 1$, so the Ratio Test says that $\displaystyle\sum_{k=0}^{\infty} \frac{4^k(k+1)}{k!}$ converges. 3

(b) We have that $\arctan(k) \geqslant \frac{\pi}{4}$ and $\sqrt{k^2+1} \leqslant \sqrt{k^2+k^2}$ for all $k \geqslant 1$. It follows that

$$0 < \frac{\pi}{4\sqrt{2}} \frac{1}{k} \leqslant \frac{\arctan(k)}{\sqrt{k^2+1}}$$

for all $k \in \mathbb{N}$. Since $\sum_{k=1}^{\infty} \frac{1}{k}$ is a divergent harmonic series, the Comparison Test says that the given series diverges. 5

4.(a) Notice that if k is a multiple of 3, then $\sin^k(k\frac{\pi}{3}) = 1$. This means that the terms of the series have a subsequence that doesn't converge to 0. The series diverges. 2

(b) Note that

$$0 \leqslant \left|(-1)^k \sin((\tfrac{1}{2})^k)\right| = \sin((\tfrac{1}{2})^k) \leqslant (\tfrac{1}{2})^k$$

for all integers $k \geqslant 0$. Since $\sum_{k=0}^{\infty} (\frac{1}{2})^k$ is a convergent geometric series, the Comparison Test tells us that $\sum_{k=0}^{\infty} \sin((\frac{1}{2})^k)$ converges. Thus $\sum_{k=0}^{\infty} (-1)^k \sin((\frac{1}{2})^k)$ is absolutely convergent. 5

(c) First note that $\left(\sqrt[3]{k^2+8}\right)_{k=0}^{\infty}$ is a strictly increasing sequence tending to ∞. It follows by the Alternating Series Test that $\sum_{k=0}^{\infty} \frac{(-1)^k}{\sqrt[3]{k^2+8}}$ converges. However, for any $k \geqslant 3$,

$$0 < \frac{1}{\sqrt[3]{2k^2}} \leqslant \frac{1}{\sqrt[3]{k^2+8}}.$$

Since $\sum_{k=1}^{\infty} \frac{1}{\sqrt[3]{k^2}}$ is a divergent p-series, the Comparison Test tells us that $\sum_{k=0}^{\infty} \frac{1}{\sqrt[3]{k^2+8}}$ diverges. Thus $\sum_{k=0}^{\infty} \frac{(-1)^k}{\sqrt[3]{k^2+8}}$ is conditionally convergent. 7

5. First note that

$$\left|\frac{f(x)}{1+f(x)} - \frac{3}{4}\right| = \frac{|f(x)-3|}{4\,|1+f(x)|}.$$

Since $\lim_{x\to 2} f(x) = 3$, given any $\varepsilon > 0$, we can find a $\delta_\varepsilon > 0$ such that

$$(\forall x \in \mathbb{R}) \qquad 0 < |x-2| < \delta_\varepsilon \implies |f(x)-3| < \varepsilon.$$

In particular, there is a $\delta_1 > 0$ such that

$$(\forall x \in \mathbb{R}) \qquad 0 < |x-2| < \delta_1 \implies |f(x)-3| < 1.$$

If $|f(x)-3| < 1$, then $2 < f(x) < 4$, so $|1+f(x)| > 3$. Given any $\varepsilon > 0$, put $\delta = \min\{\delta_1, \delta_{12\varepsilon}\}$. Then

$$(\forall x \in \mathbb{R}) \qquad 0 < |x-2| < \delta \implies \left|\frac{f(x)}{1+f(x)} - \frac{3}{4}\right| = \frac{|f(x)-3|}{4\,|1+f(x)|} < \frac{12\varepsilon}{4 \times 3} = \varepsilon,$$

6

as required.

6.(a) Let $f(x) = x^2$. This is a polynomial, so it is continuous on $\mathbb{R}$. Consider the open interval $A = (-1, 1)$. The image of A under f is the interval $[0, 1)$, which is not open. 2

(b) We need to show that for each $a \in f^{\leftarrow}(B)$, there exists an $r > 0$ such that

$$(\forall x \in \mathbb{R}) \quad |x-a| < r \implies f(x) \in B.$$

Since $a \in f^{\leftarrow}(B)$, we have that $f(a) \in B$. Because B is open, there exists an $\varepsilon > 0$ such that

$$(\forall y \in \mathbb{R}) \quad |y - f(a)| < \varepsilon \implies y \in B.$$

Since f is continuous at a, there exists a $\delta > 0$ such that

$$(\forall x \in \mathbb{R}) \quad |x-a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

Let us therefore choose $r = \delta$. If $|x-a| < r$, then $|f(x)-f(a)| < \varepsilon$, so $f(x) \in B$, as required. 7

7.(a) Given any $x \in [-r, r]$, we can find an $N \in \mathbb{N}$ so that $-\frac{1}{2} < x^k < \frac{1}{2}$ for all integers $k \geqslant N$. For such k,

$$0 \leqslant \left| \frac{x^k}{1+x^k} \right| < \frac{r^k}{1-\frac{1}{2}} = 2r^k.$$

The geometric series $\sum_{k=N}^{\infty} 2r^k$ converges, so the Weierstrass M-Test tells us that $\sum_{k=0}^{\infty} \frac{x^k}{1+x^k}$ converges absolutely and uniformly on $[-r, r]$. 5

(b) If $x = -1$, then the term $\dfrac{x^k}{1+x^k}$ is undefined for each odd value of k. 1

(c) If $x = 1$, then $\dfrac{x^k}{1+x^k} = \frac{1}{2}$ for all k, so the terms of the series don't tend to 0. 1
